

**The University of Hong Kong  
FACULTY OF ENGINEERING  
DEPARTMENT OF COMPUTER SCIENCE**

COMP7103 DATA MINING

Date: 10th December 2018

Time: 6:30pm – 8:30pm

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

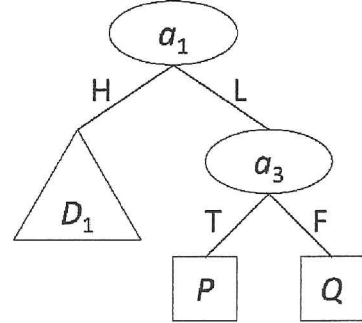
**Answer ALL 7 questions.**

1. (6%) Explain dimension attributes and measure attributes in a fact table of an OLAP system.
2. (10%) Give and explain 4 measures that evaluate the interestingness of association rules.
3. (12%) With respect to decision tree classifier construction, explain the ideas of prepruning and postpruning. Also explain the objective of tree pruning.
4. (10%) For each of the following statements, state whether the statement is TRUE or FALSE. For each false statement, point out the incorrectness.
  - (a) An association rule that satisfies a given support requirement and a given confidence requirement must have its lift larger than 1.
  - (b) The entropy of any set of labeled data is a value within the range  $[0..1]$ .
  - (c) K-means could give different clustering results with different sets of initial cluster representatives.
  - (d) All maximal frequent itemsets are closed itemsets.
  - (e) min-max normalization is better than z-score normalization in that it is less susceptible to noise and outliers.

5. Table 1 shows a dataset of 10 records, each with three attributes  $a_1$ ,  $a_2$ ,  $a_3$ , and a class label. We use the data in Table 1 to train a decision tree classifier. A partially-built decision tree is shown in Figure 1. In the figure,  $D_1$  is a subtree yet to be built and  $P$ ,  $Q$  are leaf nodes.

| ID | $a_1$ | $a_2$ | $a_3$ | class |
|----|-------|-------|-------|-------|
| 1  | H     | X     | T     | A     |
| 3  | H     | X     | F     | A     |
| 4  | H     | Y     | T     | B     |
| 5  | H     | Y     | F     | B     |
| 2  | H     | Z     | T     | A     |
| 6  | L     | X     | T     | B     |
| 7  | L     | X     | F     | B     |
| 8  | L     | Y     | T     | B     |
| 9  | L     | Y     | F     | A     |
| 10 | L     | Z     | F     | A     |

**Table 1:** A training dataset



**Figure 1:** A partially-built decision tree

- (4%) Determine the class labels of leaf nodes  $P$  and  $Q$ . Explain your answer.
- (6%) Calculate the information gain of the split at the node that uses attribute  $a_3$  as the test.
- (6%) Use classification error as the impurity measure to complete the subtree  $D_1$ . Show your steps.
- (6%) Given the test dataset shown in Table 2, calculate the F-measure for the target class “B” of the completed classification tree.

| ID | $a_1$ | $a_2$ | $a_3$ | Class |
|----|-------|-------|-------|-------|
| 11 | L     | X     | T     | A     |
| 12 | L     | X     | T     | A     |
| 13 | L     | X     | F     | B     |
| 14 | L     | Y     | T     | B     |
| 15 | L     | Y     | F     | A     |
| 16 | L     | Z     | F     | A     |

**Table 2:** A test dataset

6. Table 3 shows a market basket dataset of 10 transactions.

| Transaction ID | Items Bought |
|----------------|--------------|
| 1              | {a}          |
| 2              | {a, b, d, e} |
| 3              | {a, b, c, e} |
| 4              | {a, b, e}    |
| 5              | {a, c, e}    |
| 6              | {a, c, e}    |
| 7              | {b, c}       |
| 8              | {b, c}       |
| 9              | {b, c, d, e} |
| 10             | {b, c, e}    |

**Table 3:** A market basket dataset

Given a support threshold  $\rho_s = 25\%$  and a confidence threshold  $\rho_c = 70\%$ , answer the following questions.

- (4%) Find the support counts of all 2-itemsets.
  - (4%) List all frequent 3-itemsets.
  - (4%) List all closed frequent 2-itemsets and all maximal frequent 2-itemsets.
  - (4%) List all valid association rules (those that satisfy the confidence requirement) that are derived from the frequent 3-itemsets found.
  - (4%) Compute the lifts of the rules found in part (d).
7. Table 4 shows the values of 5 objects in a one-dimensional space.

| Object | Location |
|--------|----------|
| $O_1$  | 10       |
| $O_2$  | 11       |
| $O_3$  | 17       |
| $O_4$  | 21       |
| $O_5$  | 35       |

**Table 4:** Location of 5 objects

- (4%) Construct the proximity matrix of the objects using the L1 norm as the distance measure.
- (8%) Perform complete-link hierarchical clustering on the objects. Show your results by drawing a dendrogram. Mark the dendrogram with the proximity values of the merge points.
- (8%) Generate 2 clusters from the dendrogram in part (b), then,
  - find the centroids of the clusters;
  - calculate the SSE of the clustering, using the centroids as the cluster representatives.

END OF PAPER